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> #This code reads in the Sleep.csv data set. It then creates the sleep data frame,
> #and uses it to create a scatter plot of brain weight vs total sleep, finds the Least
> #Squares Regression Model, and then predicts total sleep when brain weight is 100 and when
> #brain weight is 500.
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> #Reads in Sleep data set, creates sleep data frame
> sleep<-read.csv("Sleep.csv")
> sleep

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	Species	Brain_Weight	Total_Sleep	Danger
1	Roe deer	98.20	2.6	5
2	Horse	655.00	2.9	5
3	Donkey	419.00	3.1	5
4	Goat	115.00	3.8	5
5	Sheep	175.00	3.8	5
6	Cow	423.00	3.9	5
7	Rock hyrax(Procaviahab)	21.00	5.4	3
8	Tree hyrax	12.30	5.4	3
9	Genet	17.50	6.1	1
10	Brazilian tapir	169.00	6.2	4
11	Gray seal	325.00	6.2	1
12	Rock hyrax(Heterob)	12.30	6.6	2
13	Guinea pig	5.50	8.2	4
14	African giant pouched rat	6.60	8.3	3
15	Eastern American mole	1.20	8.4	1
16	Pig	180.00	8.4	4
17	Rabbit	12.10	8.4	5

18	Echidna	25.00	8.6	2
19	Lesser short-tailed shrew	0.14	9.1	4
20	Rhesus monkey	179.00	9.6	2
21	Chimpanzee	440.00	9.7	1
22	Baboon	179.50	9.8	4
23	Red fox	50.40	9.8	1
24	Desert hedgehog	2.40	10.3	2
25	Starnosed mole	1.00	10.3	2
26	Vervet	58.00	10.3	4
27	Mole rat	3.00	10.6	1
28	European hedgehog	3.50	10.7	2
29	Galago	5.00	10.7	2
30	Jaguar	157.00	10.8	1
31	Patas monkey	115.00	10.9	4
32	Slow loris	12.50	11.0	2
33	Mountain beaver	8.10	11.2	3
34	Gorilla	406.00	12.0	1
35	Arctic Fox	44.50	12.5	1
36	Chinchilla	6.40	12.5	4
37	Raccoon	39.20	12.5	2
38	Musk shrew	0.33	12.8	3
39	Gray wolf	119.50	13.0	1
40	Mouse	0.40	13.2	3
41	Rat	1.90	13.2	3
42	Tenrec	2.60	13.3	2
43	Phanlanger	11.40	13.7	2
44	Ground squirrel	4.00	13.8	3
45	Golden hamster	1.00	14.4	2

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46      Cat      25.60   14.5   1
47      Tree shrew   2.50   15.8   2
48 Arctic ground squirrel   5.70   16.5   3
49      Owl monkey  15.50   17.0   2
50 Nine-banded armadillo   10.80   17.4   1
51      Giant armadillo  81.00   18.1   1
52 North American opossum   6.30   19.4   1
53      Water opossum   3.90   19.4   1
54      Big brown bat   0.30   19.7   1
55      Little brown bat  0.25   19.9   1
>
> #Create a scatter plot of brain weight vs total sleep
> ggplot(sleep, aes(x=Brain_Weight, y=Total_Sleep))+
+   geom_point()+geom_smooth(method=lm)+
+   ggtitle("Linear Relationship between Brain Weight\n
+     and Total Sleep")+
+   labs(x="Brain Weight", y="Total Sleep")+
+   theme(plot.title = element_text(hjust=.5))
`geom_smooth()` using formula = 'y ~ x'
>
> #Finding the Least Squares Regression Model
> LSRM<-lm(Total_Sleep~Brain_Weight, data=sleep)
> summary(LSRM)

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Call:

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lm(formula = Total_Sleep ~ Brain_Weight, data = sleep)
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Residuals:

```
Min 1Q Median 3Q Max
-8.0342 -2.8719 0.1718 2.0426 7.8037
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Coefficients:

```
Estimate Std. Error t value Pr(>|t|)
(Intercept) 12.099991 0.632611 19.127 < 2e-16 ***
Brain_Weight -0.014926 0.003833 -3.894 0.000278 ***
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Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.021 on 53 degrees of freedom

Multiple R-squared: 0.2224, Adjusted R-squared: 0.2078

F-statistic: 15.16 on 1 and 53 DF, p-value: 0.0002782

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> #Predicting total sleep when brain weight is 100 and when brain weight is 500

> predict(LSRM, data.frame(Brain_Weight=c(100, 500)))

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1 2
10.607375 4.636913
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